Data Cleaning

2021-07-06

[1] defines data cleaning as

> [the] process of detecting, diagnosing, and editing faulty data.

University Students Data

The field/variable names of a data set used to understand the reasons that students leave college before graduation are shown below.

- `university`: Name of university
- `gender`: Gender of the student
- `age`: Age of the student
- `mother_ed`: College experience of mother (2 year, 4 year, graduate)
- `father_ed`: College experience of father (2 year, 4 year, graduate)
- `how_funded`: How college is funded – loans, grants, scholarships, paid by family
- `mental_illness`: Mental illnesses experienced during college years (yes, no)
- `disability`: Physical disability (including pregnancy) (yes, no)
- `current_gpa`: Semester GPA in the semester when student withdrew
- `overall_gpa`: Overall GPA of the student
- `withdrew`: Did the student withdraw after current semester? (yes, no)
- `commuter`: Does the student commute? (yes, no)
- `veteran`: Is the student a veteran? (yes, no)
- `major`: Academic major(s)

Exercises

It may be helpful to review the list of potential ethical issues in computing and data science below as you consider the exercises below.

Confidentiality of personally identifying data

1. If all of the data in this survey were made available, would it be possible to identify or nearly identify some individuals (perhaps by combining it with other information)? Who might have access to the additional information that would be needed?

2. Could a personally identified individual suffer harm from being identified in these data? (What sorts of harm?)

3. What ethical responsibilities do researchers working with this data have in light of your previous answers?

Dealing with missing data

Very often some values of some variables are not recorded for some individuals. There are many reasons why data might be missing. Perhaps the data collection protocol changed over time. Perhaps some subjects declined to answer some questions on a survey. Perhaps some information is simply not available for some individuals.

Before proceeding to analyse a data set, researchers need to decide how to handle missing data.

4. For several people, `age` is missing. Consider the pros and cons of each of these possible options for addressing the problem. How likely is the option to create an ethical issue and what might that issue be if there is one?

    a. Fill with the median value from the data set.

    b. Fill with the mean value from the data set.

    c. Remove records where `age` is missing and make a note of this in the data cleaning documentation.

5. For several records, the college experience of at least one of the parents is missing. Is there a sub-population with which this might be more likely to happen? What ethical issue(s) might arise if we simply remove anyone for whom this values is missing?

6. For several records, `disability` is missing. Removing the related records could negatively affect the diversity of this study. Why?

7. There are several other fields left blank on various records. You try to take each omission on a case by case basis, doing what you think is right. This might influence both reliability and hospitality. Why?

Dealing with (possibly) incorrect data

Just because values are included in a data set doesn't mean that those values are correct. Unusual observations might catch our eye and lead us to investigate further.

8. One current GPA is listed as 1.35. This is much lower than the GPAs of other students from that university, where the mean GPA is approximately 3.0. You're pretty sure that the 1 and the 3 must have been swapped so you go ahead and change the GPA to a more reasonable 3.15. What ethical issues might you be violating?

Ethical Issues Definitions

{{% ethical-issues-list %}}

Reliability: Ensuring that one's work consistently does what claims to do.

Hospitality: Creating work that is easily understandable and usable both by peers in a similar field and by other stakeholders and interested parties.

Privacy and Security: Balancing the need for protecting personal and group data with the use of that data for the welfare of the public.

Bias: Recognizing and reducing any potential biases present in an algorithm, model, or process that may have arisen as a result of ignorance, assumptions, or past discriminatory societal patterns.

Data Integrity: Collecting and handling data in a way that accurately reflects the phenomena being studied and is appropriate for the analysis techniques employed.

Professional Ethics: Contributing one's best effort in the workplace while respecting one's peers and upholding the requirements of one's employer and the discipline's overarching guild.

References

Bibliography

[1] J. Van den Broeck, S. A. Cunningham, R. Eeckels, and K. Herbst, "Data cleaning: detecting, diagnosing, and editing data abnormalities," PLoS Med, vol. 2, no. 10, p. e267, Oct. 2005, doi: 10.1371/journal.pmed.0020267.